Nervous System Development

Learning objectives

At the end of the lecture, student should be able to;

- know the brief organization and function of nervous system
- know in depth the development of neural tube, and neural crest cells and their derivatives.
- know the details of development of various parts of brain and spinal cord.
- understand the steps of myelination.

Nervous SYSTEM –OVERVIEW

Integrate & controls of all systems of body, it is divided into 3 main parts;

The central nervous system (CNS)
includes brain & spinal cord.

The peripheral nervous system (PNS)
includes neuron out of the CNS.
Connect brain & spinal cord to outside environment.

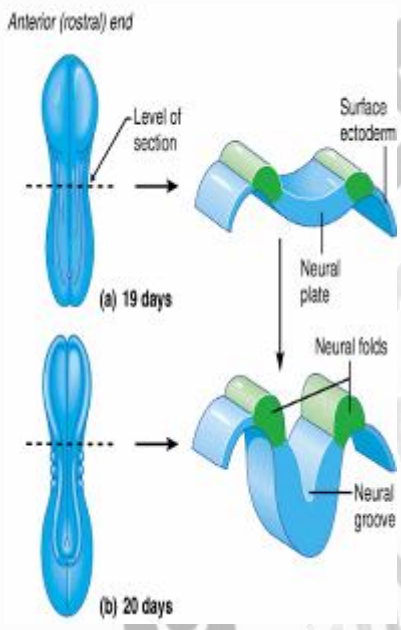
The autonomic Nervous system—(ANS)
Here the activity of this system is involuntary
Includes CNS & PNS
Innervates smooth muscle cells cardicac muscle, glandular epithelium

Development of Nervous System

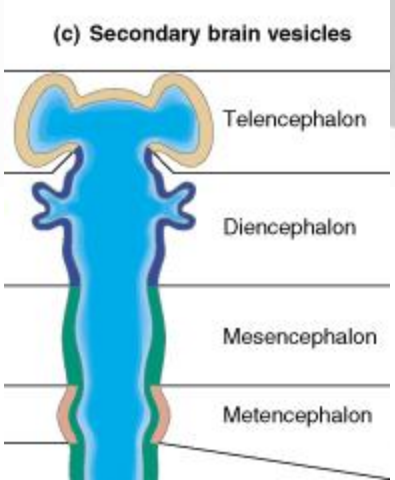
- ☒ It begins during ***third week*** of development, the ***ectoderm keep*** on proliferating &advances towards the cranial end of embryo dorsal to notochord thickened to form a plate __ ***Neural plate.***
- ☒
- ☒ All the activity occurs due to ***inductive influence of notochord.***

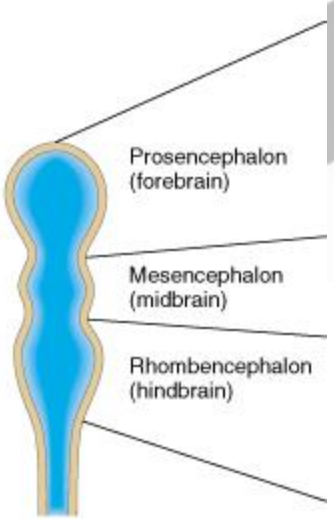
- ☒ As the notochord enlongated the neural plate broadens and extends as far as prechordal plate.

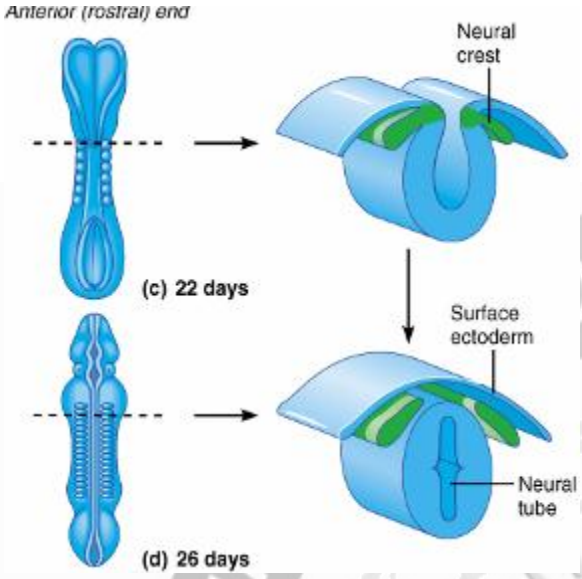
- ☒ On ***18th day*** this neural plate invaginates along central axis and deeps to form the ***Neural Groove*** in the midline.



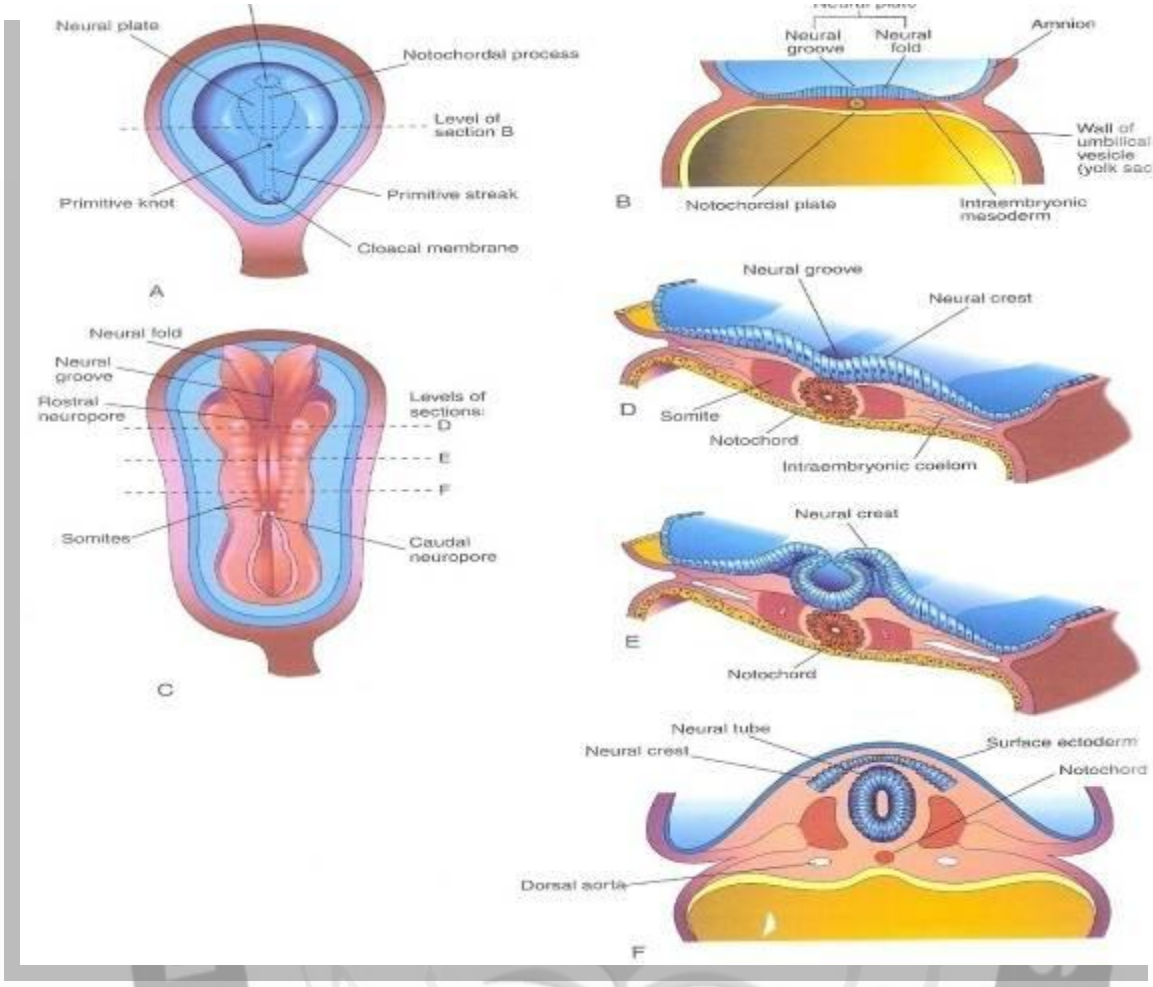
- ☒ The elevative ends on both sides the ***Neural folds*** becomes prominent.
- ☒ By the end of third week the ***neural folds*** began to move closer & grow over the neural groove forming the ***Neural tube.***
- ☒ The ***neural tube is completed by fourth week***.
- ☒ ***7. Neural tube becomes separated*** from overlying ectoderm & the ***neural crest cells*** are formed situ l tube and the surface ectoderm.
- ☒







- ☒ The neural crest cells after differentiating ,shift caudally &laterally to surround the whole extent of neural tube.
- ☒ The tube is open at both end known **anterior neuropore & posterior neuropore**
- ☒ The human of neural tube is known as neural canal & communicates freely with ammiotic cavity.
- ☒ The cranial or anterior neuropore closes at lay 25th day of development & posterior neuropore closes at day 27th day of development.
- ☒ ***Neural tube forms CNS (esp. spinal cord)***
- ☒ ***Expands anteriorly to form brain***
- ☒ ***Peripheral nerves develop from lateral "neural crest" cells***



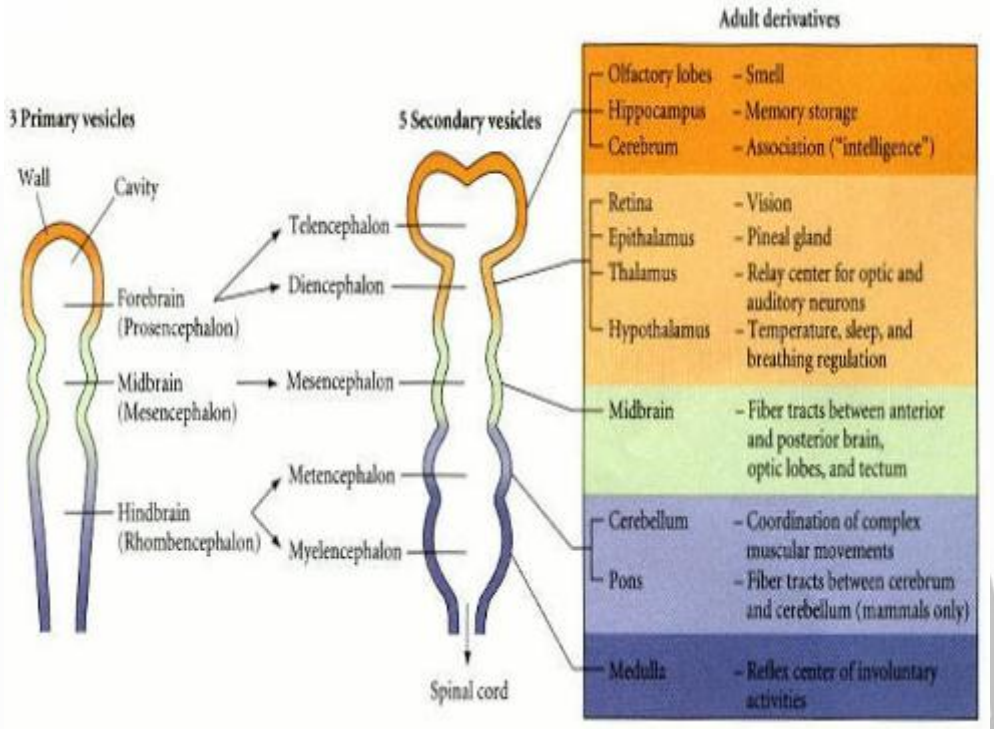
- The neural tube soon differentiate into 3 ***primary vesicle;***

Fore brain or Presencephalon.
Mid brain or Mesencephalon.
Hind brain or Rhombencephalon.

Secondary vesicles; ☒

- ☒ The procencephalon into Telencephalon- which two out pocketing on lateral aspect of forebrain.
- ☒ Rhombencephalon- metencephalon or pons
- ☒ Myelenchncephalon or medulla oblongata.
- ☒ These vesicles are communicated with each via the inter communicating canal or lumen containing CSF is known as ventricles.
- ☒ Ventricle of Telencepalon are the lateral ventricle
- ☒ The ventricle of midbrain is aqueduct or duct of sylvius
- ☒ ventricle of hind brain is fourth ventricle continuous with central canal of spinal cord.

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Adult derivatives
3 Primary vesicles
5 Secondary vesicles
Wall
Cavity
Forebrain (Prosencephalon)
Midbrain (Mesencephalon)
Hindbrain (Rhombencephalon)
Telencephalon
Diencephalon
Mesencephalon
Metencephalon
Myelencephalon
Spinal cord
Olfactory lobes – Smell
Hippocampus – Memory storage
Cerebrum – Association ("intelligence")
Retina – Vision
Epithalamus – Pineal gland
Thalamus – Relay center for optic and auditory neurons
Hypothalamus – Temperature, sleep, and breathing regulation
Midbrain – Fiber tracts between anterior and posterior brain, optic lobes, and tectum
Cerebellum – Coordination of complex muscular movements
Pons – Fiber tracts between cerebrum and cerebellum (mammals only)
Medulla – Reflex center of involuntary activities

Development of Spinal Cord

- ☒ Caudal to fourth somite stage spinal cord develops.
- ☒ Initially the wall of neural tube is made of highly specialized cells the *NEURO EPITHELIUM*.
- ☒ The ependymal cells lining the ventricles which gives rise to neurons & supporting cells.
- ☒ Ependymal cell begins to proliferate & give rise to more and more cells and by end of closure of neural tube they give rise to the *primordial neuroblast*, these form a zone extending from the ventricle known as *Mantle layer* containing the cell body or gray matter of spinal cord.
- ☒ From these cells myelinated nerve fibres extend towards the periphery forming the *marginal layer* that form the white matter of spinal cord.
- ☒ The ***neuroblasts*** gets mature & form ***adults neurons.***
- ☒ Once the neuroblast are formed they lose their ability to divide.
- ☒ ***GLIOBLASTS are primordial supporting cells***. After the neuroblast loses its ability to form neurons it then starts developing into the supporting cells.
- ☒ These glioblast migrates into Mantle & marginal zone of spinal cord.
- ☒ The main ***glioblasts*** are ***the astrocytes-*** protoplasmic fibrous ***Oligodendrocytes.***
- ☒ The ***microglial cells*** are derived from ***mesenchymal cells*** ,originate in bone marrow and are part of monocytic family

Development of Spinal Cord

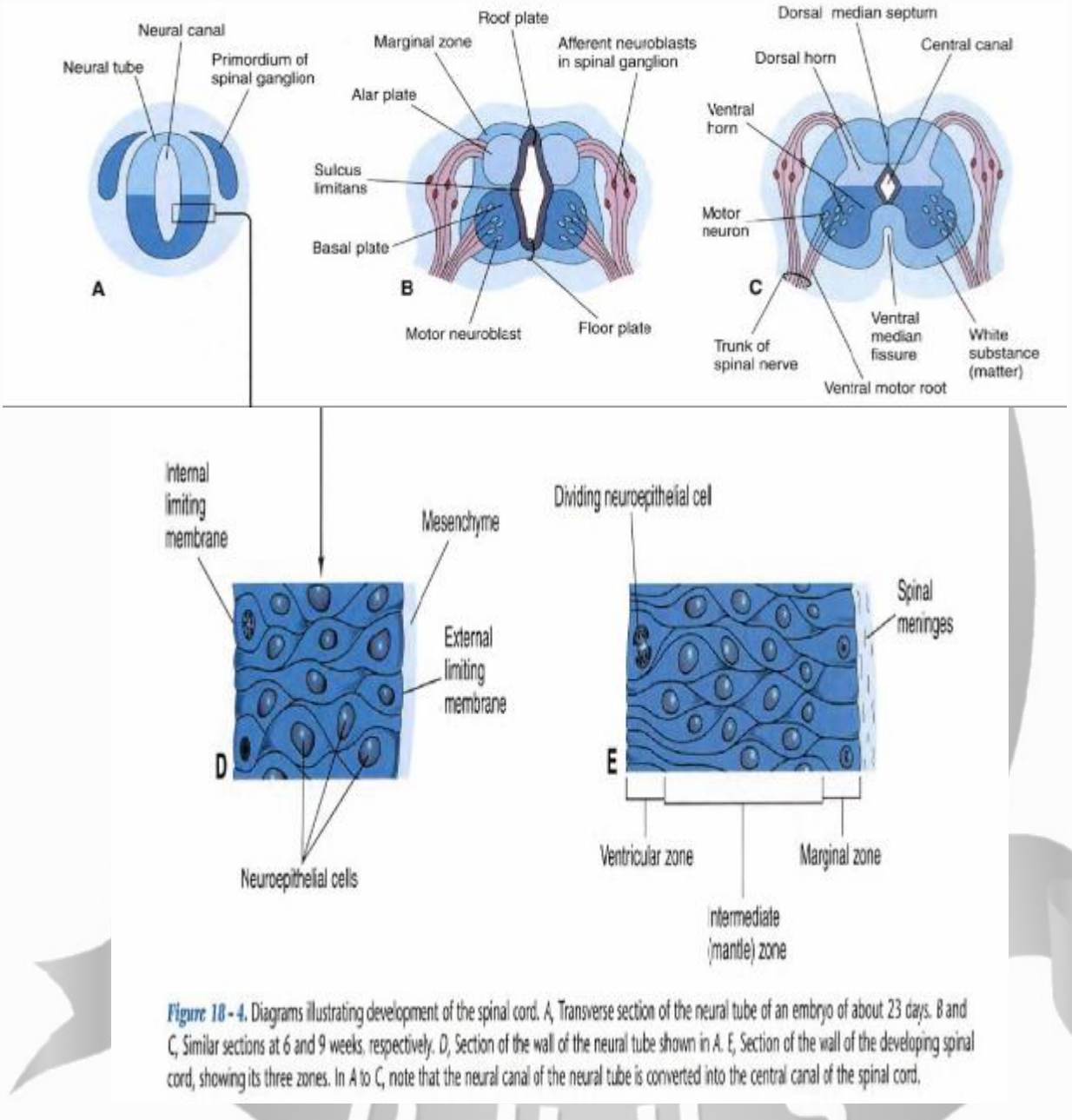
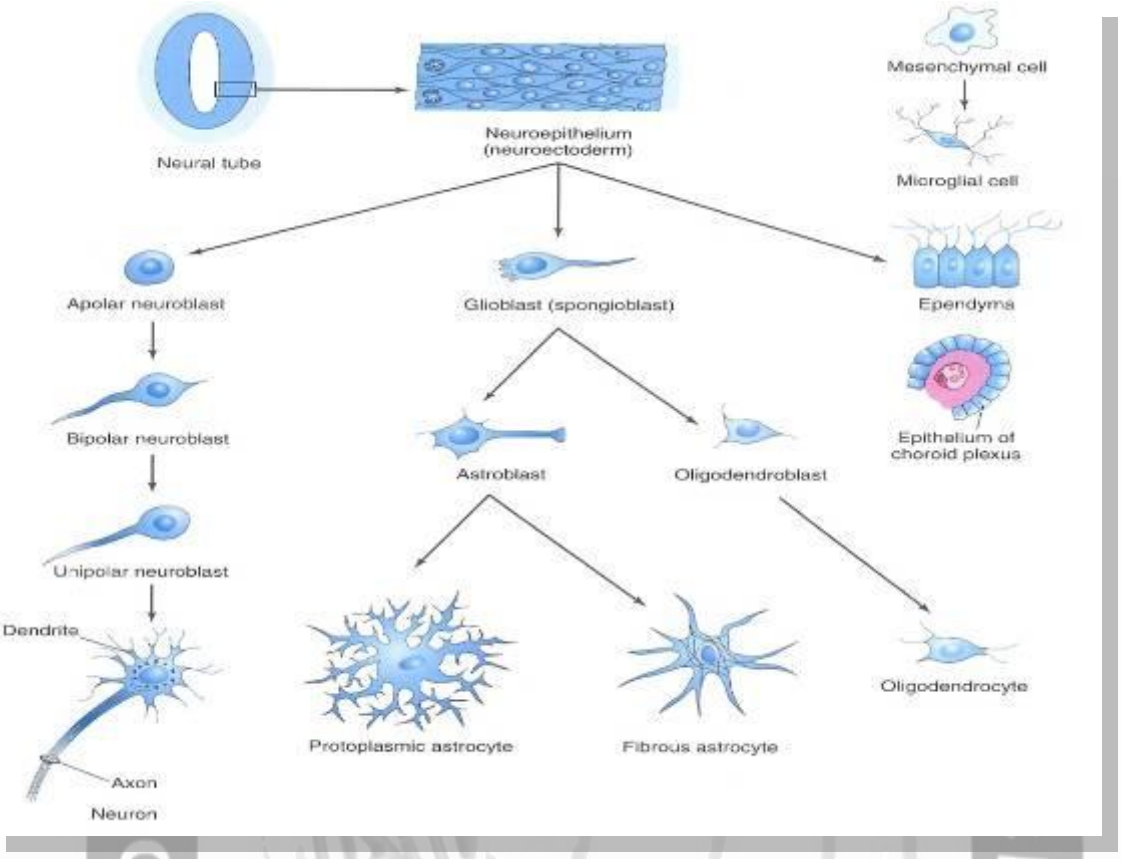


Figure 18-4. Diagrams illustrating development of the spinal cord. A, Transverse section of the neural tube of an embryo of about 23 days. B and C, Similar sections at 6 and 9 weeks, respectively. D, Section of the wall of the neural tube shown in A. E, Section of the wall of the developing spinal cord, showing its three zones. In A to C, note that the neural canal of the neural tube is converted into the central canal of the spinal cord.

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Development of Spinal Cord



The unipolar ganglia in spinal ganglion are derived from neural crest cells

This unipolar neuron gives out two process and unite in centre as "T"

The process will act as axon & other will be dendrite

The peripheral process of spinal ganglia pass in spinal nerve to sensory endings in somatic or visceral structure

The central process pass in dorsal horn & form dorsal root of spinal nerves.

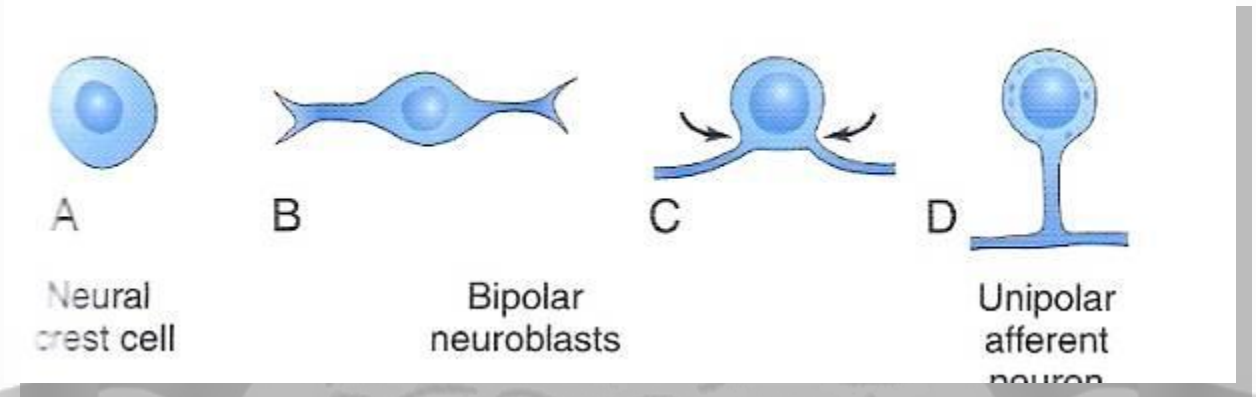
Proliferation and differentiation of the neuroepithelial cells in the developing spinal cord produce thick walls and thin roof and floor plates. A shallow longitudinal *sulcus limitans* appears in the lateral walls of the spinal cord and separates the **dorsal alar plate** from the **ventral basal plate.**

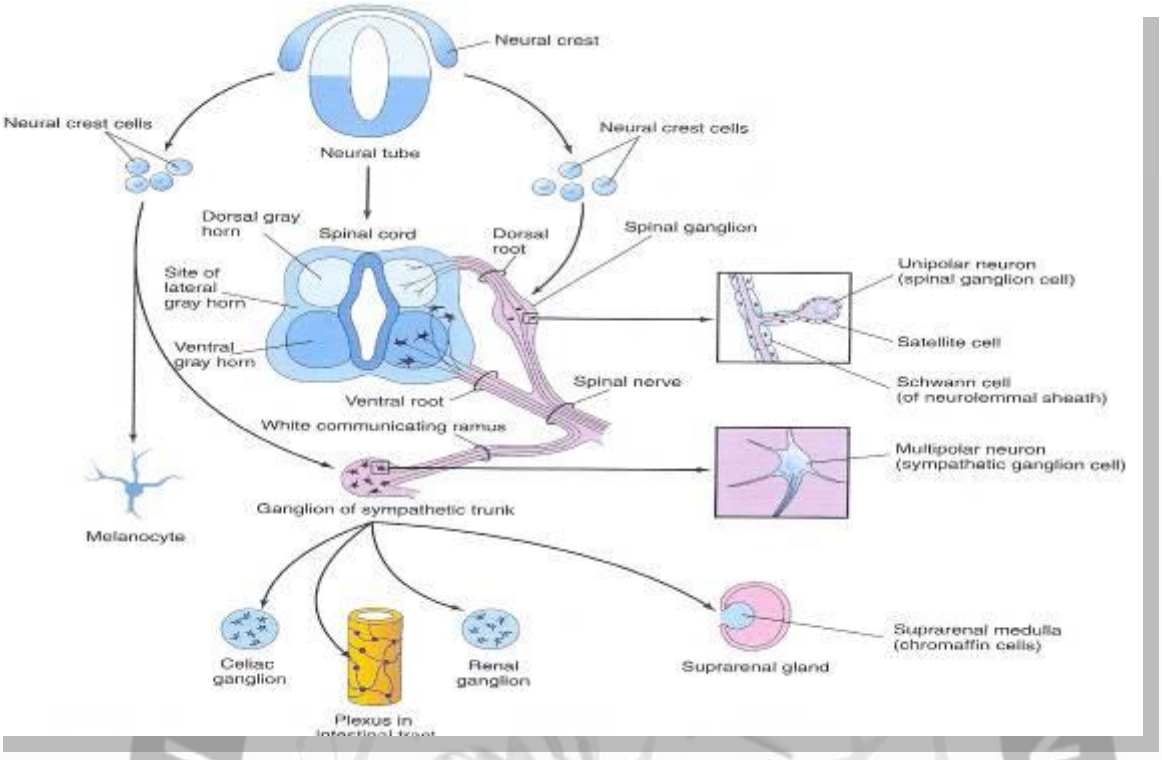
- ***Alar plates***: cells form the *dorsal horns* and will have *afferent functions.*

- ***Basal plates***: cells form the *ventral and lateral horns* and will have *efferent* functions. Axons grow out of the spinal cord to form the *ventral roots.*
- The ***dorsal root ganglia*** are formed from the *neural crest cells*. Their axons enter the spinal cord and form the *dorsal roots.*

Histological Differentiation of Cells

- ☒ When neuroblast arise it has a central process which extends to lumen—Transient dendrite—temporary dendrite & disappear, then again myoblast develops another single process in Mantle layer apolar neuroblast.
- ☒ Then cytoplasmic processes appear on both opposite sides ---- Bipolar neuroblast.
- ☒ When at one end single process (axon) is given other, then at other arborization occurs & many processes in the form of dendrites appear-- multipolar neuroblast.





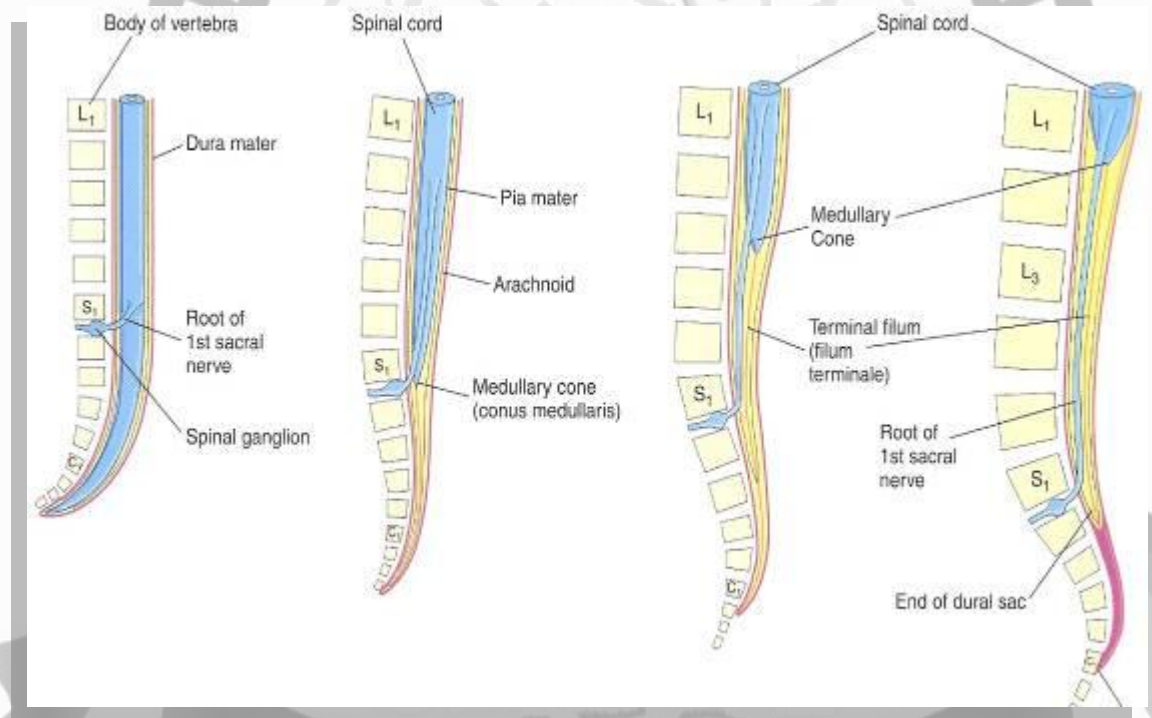
- ☒ Proliferation & differentiaiton of neuroepithelial cells form- thick wall & thin roof & floor plates, separated by a longitudinal groove & slit known as ***sulcus limitans***

- ☒ On each side of neural tube the dorsal & ventral swelling or thickening appears, which are separated by sulcus limitans

- ☒ The ventral thickening- ***basal plate*** contians ventral or anterior motor horn cells contianing afferent nuclei form ventral root of spinal vein. The dorsal thickening- **alar plate** contains dorsal horn cells or sensory areas contain efferent nuclei & forms dorsal root of spinal nerve. The lateral horn extends from T1 to L2 level.
- ☒ As the basal plate enlarge & bulgi at each side of median plane—ventral median septum is formed a deep longitudinal groove.
- ☒ Ventral median fissure develops on ventral surface of spinal cord.
- ☒

Development of Meninges

- ☒ The ***mesenchyme*** surrounding the neural tube condenses to form membrane--- *Meninges or menix.*
- ☒ The ***external layer*** is thickened to form duramater or pachyminx- derived from *ectoderm*.
- ☒ The ***internal layer*** develops into arcahnoid and piameter together known as *leptomeninges & are derived from neural crest cells.*
- ☒ The space between leptomeninges for subarchnoid space.
- ☒ Numerous delicate strands of tissues passes between the liptomeninges.
- ☒ CSF begins to form during fifth week.



Positional Change of Spinal Cord

The spinal cord in embryo

- extends the *entire length* of vertebral canal
- Spinal nerve pass out through intervertebral foramina at their level of origin
- The vertebral column & duramater lengthens rapidly then the neural tube & terminal end of spinal cord shift to higher level

At month 6 of gestation,

- end of the spinal cord lies at level *of S1*.

In the newborn infant,

- It lies at *L3*
- Spinal roots then run obliquely from their level of origin

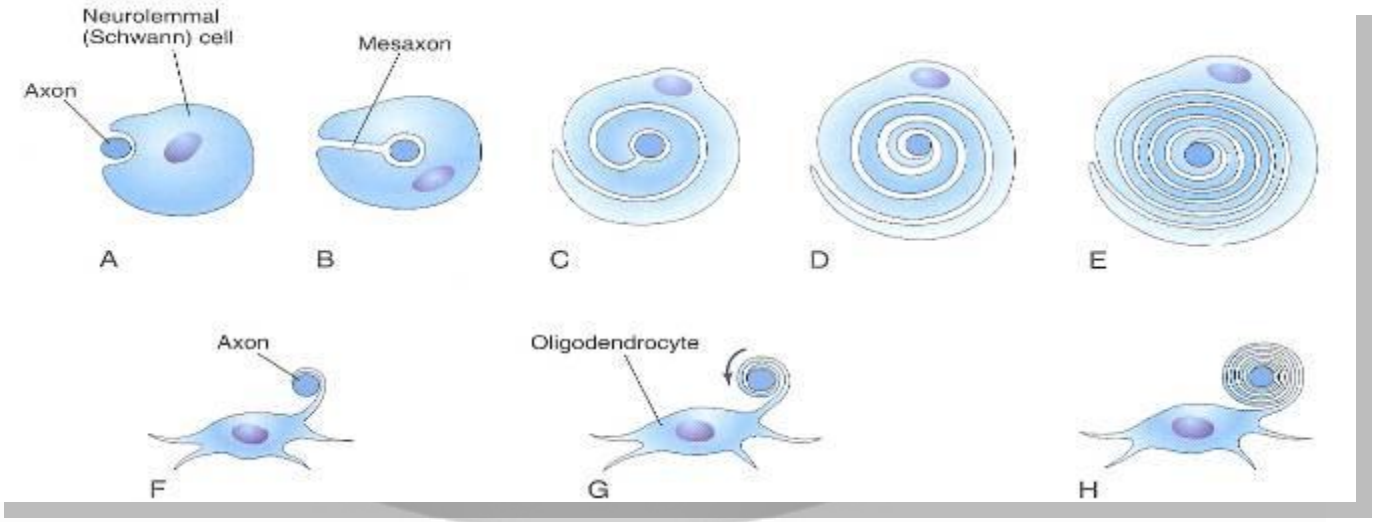
In the adult, it lies *at L 2-3*. Lumbar and sacral spinal nerve roots run obliquely from the spinal cord to their corresponding intervertebral foramina inferiorly.

- The spinal cord tapers & ends in the form of rounded swelling called ***Conus medullaris***
- Nerve roots running caudal to Conus medullaris is called ***cauda equina*** (cauda- tail, equina- horns)
- At the lower end of cord, the piamter form a tail ***Filum Terminale*** & is attached to 1st coccygeal vertebrae

Myelination of Nerve Fibers

- ☒ **A sheath is formed around nerve fibres (axon) *during the late fetal period & continues after birth till 1, ½ year.***
- ☒ **The sheath is derived from Myelin-a basic protein motor root are myelinated before sensory roots**
- ☒ **The *Schawan cells* are responsible for mylination of *peripheral nerve fibres***
- ☒ **Whereas the *oligodendrocytes* form sheath around nerve fibres in *CNS***
- ☒ **These neuroglial cells are of neural crest cell origin & wrap around the nerve fibres in several layers**

Figure 18 - 11. Diagrammatic sketches illustrating myelination of nerve fibers. *A* to *E*, Successive stages in the myelination of an axon of a peripheral nerve fiber by a neurolemmal cell. The axon first indents the cell; the cell then rotates around the axon as the mesaxon (site of invagination) elongates. The cytoplasm between the layers of the cell membrane gradually condenses. Cytoplasm remains on the inside of the sheath between the myelin and axon. *F* to *H*, Successive stages in the myelination of a nerve fiber in the CNS by an oligodendrocyte. A process of the neuroglial cell wraps itself around an axon, and the intervening layers of cytoplasm move to the body of the cell.



Points to Remember

- ☒ Nervous system develops from an area of embryonic ectoderm called the **neural plate** which appears during **week 3**
- ☒ The ***neural tube*** gives rise to the *central nervous system (brain and spinal cord)*
- ☒ The ***neural crest*** gives rise to the *peripheral nervous system* (cranial, peripheral, autonomic ganglia and nerves) and Schwann cells, pigment cells, odontoblasts, meninges, and bones and muscles of the head

Positional Change of Spinal Cord

Myelination of Nerve Fibers

Myelination of Nerve Fibers

Points to Remember

Q1.Which one is the primary brain vesical

Q3.Development of brain starts in

Q5.Name the primary brain vesicles marked as A,B and C in the diagram